PAMELLA CRISTINY CARNEIRO DA SILVA

PHYSICAL CHEMIST | COMPUTATIONAL & EXPERIMENTAL CHEMISTRY

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PROFESSIONAL PROFILE

Bridging experimental laboratory science and computational chemistry together. Physical Chemist with experience in laboratory analysis, quality management, technical responsibility, and computational chemistry. Background in physicochemical analyses, metrological traceability and ISO/IEC 17025 requirements, combined with experience in Molecular Dynamics, molecular modeling and biomolecular systems. Proficient in Julia, GROMACS, Linux and scientific data analysis. Interested in pharmaceutical, biotechnology, analytical, R&D and computational chemistry roles.

CORE EXPERTISE

Laboratory & Quality: Physicochemical Analysis · Microbiological Testing · Quality Control · ISO/IEC 17025 · Metrological Traceability · Technical Responsibility · Sampling protocols

Computational Chemistry: Molecular Dynamics · Molecular Modeling · GROMACS · Julia · Linux · Scientific Data Analysis

Biomolecular Applications: Proteins · Antibodies · Biomolecular Systems · Medicinal Chemistry · Solvation Effects

PROFESSIONAL EXPERIENCE

Technical Manager / Laboratory Analyst | LAS Sanches Laboratório / *Apr 2018 – Apr 2021* | Brazil

- Technical responsibility for water and industrial wastewater testing and laboratory activities.
- Performed physicochemical analyses including pH, conductivity, turbidity, COD/BOD, solids, alkalinity, hardness, chlorine, titration and spectrophotometry.
- Performed microbiological testing for coliforms and Escherichia coli; operated spectrophotometers and Gallery automated analyzer.
- Managed laboratory quality procedures, analytical records and metrological traceability.
- Applied ABNT NBR ISO/IEC 17025 requirements and evaluated analytical results and test conformity.

Guest Researcher | Delft University of Technology (TU Delft) / *Mar 2026 – Present* | Delft, Netherlands

- Molecular Dynamics simulations and computational analysis of molecular and biomolecular systems.
- Developed computational workflows using GROMACS and Julia for physicochemical and trajectory analysis.
- Investigated protein–solvent interactions and molecular properties relevant to pharmaceutical and biomolecular research.

Research Intern — Artificial Intelligence | University of São Paulo (EACH-USP) / *2016* | São Paulo, Brazil

- Research experience in artificial intelligence and computational approaches applied to scientific problems.
- Scientific programming, computational analysis and research activities in an AI laboratory.

TECHNICAL SKILLS

Simulation & Software: GROMACS · PACKMOL · PLUMED · Molecular Dynamics workflows · Trajectory analysis

Programming & Data: Julia · Linux/Unix · Scientific programming · Data processing · Data visualization

Laboratory Methods: pH · Conductivity · Turbidity · COD/BOD · Solids · Alkalinity · Hardness · Chlorine · Titration · Metals · Spectrophotometry · Microbiological testing

Instrumentation: Spectrophotometers · Gallery automated analyzer

Quality: ABNT NBR ISO/IEC 17025 · Metrological traceability · Quality control · Laboratory documentation

EDUCATION

PhD Candidate in Physical Chemistry | University of Campinas (UNICAMP) / *Feb 2024 – Present* | Brazil

Research: Computational chemistry, Molecular Dynamics, solvation, complex systems, biomolecular interactions, Ionic Liquids and vaccine adjuvants.

MSc in Physical Chemistry | University of Campinas (UNICAMP) / *Jan 2021 – Oct 2023* | Brazil

Research: Molecular modeling, Molecular Dynamics, antigen-antibody interactions for medicinal chemistry.

BSc in Industrial Chemistry | Federal University of Pará (UFPA) / *Mar 2014 – Mar 2018* | Belém, Brazil

ADDITIONAL RESEARCH EXPERIENCE

Scientific Initiation — Molecular Modeling Laboratory | Federal University of Pará (UFPA) / *Jan 2015 – Mar 2017* | Belém, Brazil

- Research experience in medicinal chemistry through a scholarship scientific initiation program.

SELECTED SCIENTIFIC OUTPUTS

- Extended Conformational Selection in the Antigen–Antibody Interaction of the PFAMA1 Protein — [Journal of Physical Chemistry B / 2024 / https://doi.org/10.1021/acs.jpcb.4c03734]
- Intrinsically Disordered Malaria Antigens: An Overview of Structures, Dynamics and Molecular Simulation Opportunities — [Journal of Brazilian Chemical Society / 2024 / https://jbcs.sbq.org.br/pdf/2024-0023RV_MedChem]
- Molecular Dynamics Tutorial using GROMACS & PACKMOL — GitHub: [https://github.com/pamellaccar]

AWARDS

- Highlight in Scientific and Technological Initiation — Federal University of Pará (UFPA), 2016.
- Scientific Initiation Summer Scholarship — Artificial Intelligence Laboratory, University of São Paulo (EACH-USP).

LANGUAGES

Portuguese: Native | English |